

FD-X Dryer Controller SPI Communications Polls / Commands

The FD-X PCT/DHD dryer controller implements version 4.0 of the SPI protocol. The controller supports all required polls and commands for 'generic' SPI dryers, plus many polls and commands added as 'vendor-unique' messaging.

The SPI Device Identification Code (DEVID) for the FD-X PCT/DHD dryer controller is 0x22 (SPI dryer).

All multi-byte, numeric values are represented (transmitted) in 'big-endian' format with the most significant byte transmitted first.

SPI "Generic" Dryer Commands/Polls

1. SPI "Echo" Poll

CMD1: 0x20

CMD2: 0x20

Data: 4-byte ASCII data, returns exact data provided through the "Echo" command

2. SPI "Echo" Command

CMD1: 0x20

CMD2: 0x21

Data: 4-byte ASCII data. The recipient must retain the command data and return via the Echo Poll

3. **SPI Version Poll**

CMD1: 0x20

CMD2: 0x22

Data: 4-byte ASCII data, defines the SPI protocol version supported by the controller.

Byte #0: '0'

Byte #1: '4'

Byte #2: '0'

Byte #3: '0'

4. **SPI Process Heater #1 Setpoint Temperature Poll/Command**

CMD1: 0x20

CMD2: 0x30 (Poll)

0x31 (Command)

Data: 4-byte floating point data, in °F

5. **SPI Process Heater #1, High Process Temperature Alarm 'Delta' Temp Poll/Command**

CMD1: 0x20

CMD2: 0x32 (Poll)

0x33 (Command)

Data: 4-byte floating point data, in °F

6. SPI Process Heater #1, Low Process Temperature Alarm 'Deviation' Temp Poll/Command

CMD1: 0x20

CMD2: 0x34 (Poll)
0x35 (Command)

Data: 4-byte floating point data, in °F

Notes: The FD-X controller does not use a 'delta' or 'deviation' temperature in its process undertemp alarm testing, but this poll/command pair is required by the SPI specification. The controller will respond correctly to both messages, but does not actually implement the command. The poll response always is 0.0.

7. SPI Process Status Poll

CMD1: 0x20

CMD2: 0x40

Data: 2-byte bitmapped data

Byte #0

Bit 0: 'Processing'

1 = the dryer is in normal operation

0 = the dryer is in normal shutdown, alarm shutdown, or another operating mode

Bit 1: System Alarm

1 = the dryer has at least one alarm currently posted

0 = the dryer is not in alarm

Bit 2: Process Alarm

1 = the dryer has at least one 'process-related' alarm currently posted

0 = the dryer has no 'process-related' alarms posted

Bit 3: Machine Alarm

1 = the dryer has at least one 'machine-related' alarm currently posted

0 = the dryer has no 'machine-related' alarms posted

Bit 4: High Temp Alarm

1 = the dryer has a process or regen heater overtemp alarm currently posted

0 = the dryer has no high temp alarm posted

Bit 5: Low Temp Alarm

1 = the dryer has a process or regen heater undertemp alarm currently posted

0 = the dryer has no low temp alarm posted

Bit 6: Reserved

Bit 7: Reserved

Byte #1

Bit 0: 'Clogged Filter' Alarm

1 = the dryer has its 'dirty filter' alarm posted

0 = the dryer has no related alarms posted

Bit 1: Low Flow Alarm – not implemented, always '0'

Bit 2: High Dewpoint Alarm

1 = the dryer has its high process dewpoint alarm currently posted

0 = the dryer has no related alarms posted

Bit 3: Reserved

Bit 4: Reserved

Bit 5: Reserved

Bit 6: Reserved

Bit 7: Reserved

8. SPI Machine Status 1 Poll

CMD1: 0x20

CMD2: 0x42

Data: 2-byte bitmapped data

Byte #0

Bit 0: 'Processing'

1 = the dryer is in normal operation

0 = the dryer is in normal shutdown, alarm shutdown, or another operating mode

Bit 1: System Alarm

1 = the dryer has at least one alarm currently posted

0 = the dryer is not in alarm

Bit 2: Process Alarm

1 = the dryer has at least one 'process-related' alarm currently posted

0 = the dryer has no 'process-related' alarms posted

Bit 3: Machine Alarm

1 = the dryer has at least one 'machine-related' alarm currently posted

0 = the dryer has no 'machine-related' alarms posted

Bit 4: High Temp 'Safety Limit' Alarm – not implemented, always '0'

Bit 5: Low Temp 'Safety Limit' Alarm – not implemented, always '0'

Bit 6: Reserved

Bit 7: Reserved

Byte #1

Bit 0: Reserved

Bit 1: Low Flow Alarm – not implemented, always '0'

Bit 2: High Voltage Alarm – not implemented, always '0'

Bit 3: Low Voltage Alarm – not implemented, always '0'

Bit 4: High Current Alarm – not implemented, always '0'

Bit 5: Low Current Alarm – not implemented, always '0'

Bit 6: Phase Alarm – not implemented, always '0'

Bit 7: Reserved

9. SPI Machine Status 2 Poll

CMD1: 0x20

CMD2: 0x44

Data: 2-byte bitmapped data

Byte #0

Bit 0: 'Processing'

1 = the dryer is in normal operation

0 = the dryer is in normal shutdown, alarm shutdown, or another operating mode

Bit 1: System Alarm

1 = the dryer has at least one alarm currently posted

0 = the dryer is not in alarm

Bit 2: Process Alarm

1 = the dryer has at least one 'process-related' alarm currently posted

0 = the dryer has no 'process-related' alarms posted

Bit 3: Machine Alarm

1 = the dryer has at least one 'machine-related' alarm currently posted

0 = the dryer has no 'machine-related' alarms posted

Bit 4: Sensor Fault

1 = the dryer has at least one 'sensor fault' alarm currently posted

0 = the dryer has no related alarms posted

Bit 5: Calibration Fault – not implemented, always '0'

Bit 6: Reserved

Bit 7: Reserved

Byte #1

Bit 0: Reserved
Bit 1: Reserved
Bit 2: Reserved
Bit 3: Reserved
Bit 4: Reserved
Bit 5: Reserved
Bit 6: Reserved
Bit 7: Reserved

10. **SPI Machine Mode Poll/Command**

CMD1: 0x20

CMD2: 0x48 (Poll)
0x49 (Command)

Data: 2-byte bitmapped data

Byte #0

Bit 0: Machine On/Off
Poll: 1 = dryer is in operation, 0 = dryer is in shutdown
Command: 1 = dryer start command, 0 = dryer shutdown command

Bit 1: Alarm Acknowledge, unused

Bit 2: Reserved
Bit 3: Reserved
Bit 4: Reserved
Bit 5: Reserved
Bit 6: Reserved
Bit 7: Reserved

Byte #1

Bit 0: Reserved
Bit 1: Reserved
Bit 2: Reserved
Bit 3: Reserved
Bit 4: Reserved
Bit 5: Reserved
Bit 6: Reserved
Bit 7: Reserved

11. SPI “Protected” Machine Mode Poll/Command

CMD1: 0x20

CMD2: 0x4A (Poll)
0x4B (Command)

Byte #0

Bit 0: Machine On/Off

Poll: 1 = dryer is in operation, 0 = dryer is in shutdown

Command: 1 = dryer start command, 0 = dryer shutdown command

Bit 1: On/Off Action Allowed

Poll: 1 = always returns 0

Command: 1 = command action allowed, 0 = command action inhibited

Bit 2: Alarm Acknowledge, unused

Bit 3: Reserved

Bit 4: Reserved

Bit 5: Reserved

Bit 6: Reserved

Bit 7: Reserved

Byte #1

Bit 0: Reserved
Bit 1: Reserved
Bit 2: Reserved
Bit 3: Reserved
Bit 4: Reserved
Bit 5: Reserved
Bit 6: Reserved
Bit 7: Reserved

12. SPI Process Heater #1, Process Temperature Poll

CMD1: 0x20

CMD2: 0x70

Data: 4-byte floating point data, in °F

13. SPI Process Heater #1, Hopper Return Temperature Poll

CMD1: 0x20

CMD2: 0x72

Data: 4-byte floating point data, in °F

14. SPI Process Flow Poll

CMD1: 0x20

CMD2: 0x76

Data: 4-byte floating point data, in CFM

Notes: This poll returns valid data if the dryer is equipped with the optional 4-20 mA process airflow sensor. If the sensor is not installed, this poll always returns 0 CFM.

15. SPI Process Dewpoint Poll

CMD1: 0x20

CMD2: 0x7C

Data: 4-byte floating point data, in °F

Notes: This poll returns valid data if the dryer is equipped with a dewpoint sensor. If the sensor is not installed, this poll always returns 0 °F.

16. SPI High Dewpoint Alarm Setpoint Poll/Command

CMD1: 0x20

CMD2: 0x80 (Poll)
0x81 (Command)

Data: 4-byte floating point data, in °F

17. SPI Blanket Poll 1

CMD1: 0x20

CMD2: 0xE0

Data: multi-byte data structure

Byte #0: 0x20

Byte #1: 0x30

Bytes #2-5: Process Heater #1 Setpoint Temperature, 4-byte floating point, in °F

Byte #6: 0x20

Byte #7: 0x32

Bytes #8-11: Process Heater #1 High Process Temperature Alarm 'Delta' Temp, 4-byte floating point, in °F

Byte #12: 0x20

Byte #13: 0x30

Bytes #14-17: all '0'

Byte #18: 0x20
Byte #19: 0x40
Bytes #20-21: 2-byte bitmapped Process Status data (see earlier description)

Byte #22: 0x20
Byte #23: 0x40
Bytes #24-27: Process Heater #1, Process Temperature, 4-byte floating point, in °F

UnaDyn 'Vendor-Unique' FD-X Dryer Commands/Polls

18. ID Poll

CMD1: 0xC2
CMD2: 0x28

Data: multi-byte data structure

Byte #0: Poll response version (= 3)
Bytes #1-2: Machine software version (2-byte unsigned integer)
(implied decimal point , i.e. version "1.00" transmitted as "100")

Byte #3

Bit 0: 1 = I/O expansion card installed, 0 = no expansion card
Bit 1: 1 = PCT regeneration, 0 = 'standard' regeneration
Bit 2: 1 = Process heater #1 installed, 0 = no process heater #1
Bit 3: 1 = Process heater #1 is gas, 0 = process heater #1 is electric
Bit 4: 1 = Gas process heater #1 supports 'high turndown' ops, 0 = no high-turndown
Bit 5: 1 = Process heater #2 installed, 0 = no process heater #2
Bit 6: 1 = Regen heater is gas, 0 = regen heater is electric
Bit 7: 1 = Process blower VFD installed, 0 = no VFD

Byte #4

- Bit 0: 1 = 4-20 mA airflow sensor installed, 0 = no sensor
- Bit 1: 1 = Pressure sensor #1 installed, 0 = no sensor
- Bit 2: 1 = Pressure #1 is hopper differential, 0 = pitot tube
- Bit 3: spare
- Bit 4: spare
- Bit 5: 1 = Pressure sensor #2 installed, 0 = no sensor
- Bit 6: 1 = Pressure sensor #3 installed, 0 = no sensor
- Bit 7: 1 = Return dewpoint sensor installed, 0 = no sensor

Byte #5

- Bit 0: 1 = Cooling coil installed, 0 = no cooling coil
- Bit 1: 1 = Loss of process airflow snap switch installed, 0 = no sensor
- Bit 2: 1 = Dirty filter snap switch installed, 0 = no sensor
- Bit 3: 1 = Hopper #1 level sensor installed, 0 = no sensor
- Bit 4: 1 = Hopper #2 level sensor installed, 0 = no sensor
- Bit 5: 1 = VCL vacuum breaker installed, 0 = no vacuum breaker
- Bit 6: 1 = VCL segregation valve installed, 0 = no seg valve
- Bit 7: 1 = VCL filter pulse installed, 0 = no pulse

Byte #6

Bits 0-3: Process dewpoint sensor type

0 = no sensor

1 = 'standard' sensor

2 = Kahn Transmet sensor (10-bit A/D)

3 = Kahn Transmet sensor (12-bit A/D)

- Bit 4: 1 = 4-20 mA analog hopper level sensor installed, 0 = no sensor
- Bit 5: spare
- Bit 6: spare
- Bit 7: spare

Byte #7: PCB Revision

19. “Ops Status” Poll

CMD1: 0xC2

CMD2: 0x2A

Data: multi-byte data structure

Byte #0: Poll response version (= 1)

Byte #1: Current ‘ops’ state
0 = prestart
1 = normal ops
2 = normal shutdown
3 = alarm shutdown
4 – outputs testing
5 – hopper fill

Byte #2: Current regeneration ‘cam’

Bytes #3-4: Seconds remaining in the current cam (2-byte unsigned integer)

Byte #5

Bit 0: 1 = In ‘cooldown’

Bit 1: 1 = At least 1 active alarm posted, 0 = no alarms

Bit 2: 1 = Process heater #1 running its Material Saver setpoint, 0 = regular setpoint

Bit 3: 1 = Process heater #2 running its Material Saver setpoint, 0 = regular setpoint

Bit 4: 1 = spare

Bit 5: 1 = spare

Bit 6: 1 = spare

Bit 7: 1 = spare

Byte #6: Current process blower VFD output percentage

20. Alarms Status Poll

CMD1: 0xC2

CMD2: 0x2C

Data: multi-byte data structure

Byte #0: Poll response version (= 3)

Byte #1

- Bit 0: 1 = Process heater #1 overtemp alarm posted
- Bit 1: 1 = Process heater #1 undertemp alarm posted
- Bit 2: 1 = Process heater #1 gas controller fault alarm posted
- Bit 3: 1 = Process heater #1 contactor life notice alarm posted
- Bit 4: 1 = Process heater #2 overtemp alarm posted
- Bit 5: 1 = Process heater #2 undertemp alarm posted
- Bit 6: 1 = Process heater #2 contactor life notice alarm posted
- Bit 7: 1 = spare

Byte #2

- Bit 0: 1 = Regen heater overtemp alarm posted
- Bit 1: 1 = Regen heater undertemp alarm posted
- Bit 2: 1 = Regen heater gas controller fault alarm posted
- Bit 3: 1 = Regen heater contactor life notice alarm posted
- Bit 4: 1 = High process dewpoint alarm posted
- Bit 5: 1 = Process dewpoint sensor fail alarm posted
- Bit 6: 1 = Return dewpoint sensor fail alarm posted
- Bit 7: 1 = Regen Isolation Valve, Position Proof alarm posted

Byte #3

- Bit 0: 1 = Left bed note alarm posted
- Bit 1: 1 = Right bed note alarm posted
- Bit 2: 1 = High regen filter temp alarm posted
- Bit 3: 1 = Hopper #1 low material level alarm posted
- Bit 4: 1 = Hopper #2 low material level alarm posted
- Bit 5: 1 = Main PCB clock error alarm posted
- Bit 6: 1 = Process blower VFD fault alarm posted
- Bit 7: 1 = "Loss of compressed air" alarm posted

Byte #4

- Bit 0: 1 = Low process airflow alarm posted
- Bit 1: 1 = High process airflow alarm posted

- Bit 2: 1 = Process airflow sensor fail alarm posted
- Bit 3: 1 = Low process airflow (discrete safety switch) alarm posted
- Bit 4: 1 = Low process airflow pressure (pitot) alarm posted
- Bit 5: 1 = High process airflow pressure (pitot) alarm posted
- Bit 6: 1 = Low hopper differential pressure alarm posted
- Bit 7: 1 = High hopper differential pressure alarm posted

Byte #5

- Bit 0: 1 = Low process filter pressure alarm posted
- Bit 1: 1 = High process filter pressure alarm posted
- Bit 2: 1 = Low regen filter pressure alarm posted
- Bit 3: 1 = High regen filter pressure alarm posted
- Bit 4: 1 = Dirty filter (discrete safety switch) alarm posted
- Bit 5: 1 = VCL chamber A high vacuum alarm posted
- Bit 6: 1 = VCL chamber B high vacuum alarm posted
- Bit 7: 1 = Low hopper material level (analog level sensor) alarm posted

Byte #6

- Bit 0: 1 = Hardware fault, no analog PCB found alarm posted
- Bit 1: 1 = Hardware fault, I/O data path fail alarm posted
- Bit 2: 1 = Hardware fault, analog PCB 24 VDC fail alarm posted
- Bit 3: 1 = Hardware fault, main PCB 24 VDC fail alarm posted
- Bit 4: 1 = Hardware fault, main PCB 5 VDC low alarm posted
- Bit 5: 1 = Hardware fault, main PCB 5 VDC high alarm posted
- Bit 6: 1 = Hardware fault, I/O expansion PCB blow fuse alarm posted
- Bit 7: 1 = spare

Byte #7

- Bit 0: 1 = Heater selector switch fail alarm posted
- Bit 1: 1 = Heater selector switch 'invalid throw' alarm posted
- Bit 2: 1 = Analog level load fault alarm posted
- Bit 3: 1 = Analog level sensor fail alarm posted
- Bit 4: 1 = spare
- Bit 5: 1 = spare
- Bit 6: 1 = spare
- Bit 7: 1 = spare

21. Clear Alarms Command

CMD1: 0xC2

CMD2: 0x2D

Data: None

22. Analogs Poll

CMD1: 0xC2

CMD2: 0x2E

Data: multi-byte data structure

Byte #0: Poll response version (= 2)

Bytes #1-2: Process heater #1 temp, in °F (2-byte signed integer)

Bytes #3-4: Hopper #1 return temp, in °F (2-byte signed integer)

Bytes #5-6: Process heater #2 temp, in °F (2-byte signed integer)

Bytes #7-8: Hopper #2 return temp, in °F (2-byte signed integer)

Bytes #9-10: Regen heater temp, in °F (2-byte signed integer)

Bytes #11-12: Regen outlet temp, in °F (2-byte signed integer)

Bytes #13-14: Dryer inlet temp, in °F (2-byte signed integer)

Bytes #15-16: Throat temp, in °F (2-byte signed integer)

Bytes #17-18: Left bed desiccant temp, in °F (2-byte signed integer)

Bytes #19-20: Hopper temp #1, in °F (2-byte signed integer)

Bytes #21-22: Hopper temp #2, in °F (2-byte signed integer)

Bytes #23-24: Hopper temp #3, in °F (2-byte signed integer)

Bytes #25-26: Hopper temp #4, in °F (2-byte signed integer)

Bytes #27-28: Hopper temp #5, in °F (2-byte signed integer)

Bytes #29-30: Hopper temp #6, in °F (2-byte signed integer)

Bytes #31-32: Hopper temp #7, in °F (2-byte signed integer)

Bytes #33-34: Process dewpoint temp, in °F (2-byte signed integer)

Bytes #35-36: Return dewpoint temp, in °F (2-byte signed integer)

Bytes #37-40: Process airflow, in CFM (4-byte floating point)
Bytes #41-44: Process airflow, in FPM (4-byte floating point)
Bytes #45-48: Pressure sensor #1 value, in inches/H2O (4-byte floating point)
Bytes #49-52: Pressure sensor #2 value, in inches/H2O (4-byte floating point)
Bytes #53-56: Pressure sensor #3 value, in inches/H2O (4-byte floating point)
Bytes #57-60: Analog level sensor value (4-byte floating point)

23. Process Heater #2 Setpoint Temperature Poll/Command

CMD1: 0xC2
CMD2: 0x30 (Poll)
0x31 (Command)

Data: 4-byte floating point data, in °F

24. Process Heater #2, High Process Temperature Alarm 'Delta' Temp Poll/Command

CMD1: 0xC2
CMD2: 0x32 (Poll)
0x33 (Command)

Data: 4-byte floating point data, in °F

25. Process Heater #1, Material Saver Settings Poll/Command

CMD1: 0xC2

CMD2: 0x34 (Poll)
0x35 (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1: 1 = Material Saver enabled, 0 = disabled

Bytes #2-3: Material Saver setpoint temp, in °F (2-byte unsigned integer)

Bytes #4-5: Material Saver ΔT_1 temp, in °F (2-byte unsigned integer)

Bytes #6-7: Material Saver ΔT_2 temp, in °F (2-byte unsigned integer)

26. Process Heater #2, Material Saver Settings Poll/Command

CMD1: 0xC2

CMD2: 0x36 (Poll)
0x37 (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1: 1 = Material Saver enabled, 0 = disabled

Bytes #2-3: Material Saver setpoint temp, in °F (2-byte unsigned integer)

Bytes #4-5: Material Saver ΔT_1 temp, in °F (2-byte unsigned integer)

Bytes #6-7: Material Saver ΔT_2 temp, in °F (2-byte unsigned integer)

27. Process Heater #1 Material Saver Reset Command

CMD1: 0xC2

CMD2: 0x39

Data: None

28. Process Heater #2 Material Saver Reset Command

CMD1: 0xC2

CMD2: 0x3B

Data: None

29. Dewpoint Extend Settings Poll/Command

CMD1: 0xC2

CMD2: 0x3C (Poll)
0x3D (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1: 1 = Dewpoint Extend enabled, 0 = disabled

Bytes #2-3: Exit dewpoint temp, in °F (2-byte signed integer)

Bytes #4-5: Maximum duration, in hours (2-byte unsigned integer)

30. Get/Set PCB Clock (Poll/Command)

CMD1: 0xC2

CMD2: 0x3E (Poll)
0x3F (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1: Day (Sun = 1, Mon = 2, ..., Sat = 7)

Byte #2: Month (Jan = 1, Feb = 2, ..., Dec = 12)

Byte #3: Date (1..31)

Byte #4: Year (0..99)

Byte #5: Hour (0..23)

Byte #6: Minute (0..59)

31. Cooling Coil Ops Settings Poll/Command

CMD1: 0xC2

CMD2: 0x40 (Poll)
0x41 (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Bytes #1-2: Control setpoint temp, in °F (2-byte unsigned integer)

Bytes #3-4: Hysteresis control delta temp, in °F (2-byte signed integer)

32. AEO Blower Ops Settings Poll/Command

CMD1: 0xC2

CMD2: 0x42 (Poll)
0x43 (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1: 1 = AEO ops enabled, 0 = disabled

Bytes #2-3: Control setpoint temp, in °F (2-byte unsigned integer)

Bytes #4-5: Deadband temp, in °F (2-byte unsigned integer)

Bytes #6-7: Start delay, in minutes (2-byte unsigned integer)

Bytes #8-9: Update rate, in minutes (2-byte unsigned integer)

Bytes #10-11: Maximum percentage change, per update (2-byte unsigned integer)

Bytes #12-13: Minimum percentage output allowed (2-byte unsigned integer)

33. Heater Cooldown Settings Poll/Command

CMD1: 0xC2

CMD2: 0x46 (Poll)
0x47 (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Bytes #1-2: Maximum duration, in seconds (2-byte unsigned integer)

Bytes #3-4: Exit temp, in °F (2-byte unsigned integer)

34. VCL Settings Poll/Command

CMD1: 0xC2

CMD2: 0x4A (Poll)
0x4B (Command)

Data: multi-byte data structure

Byte #0: Poll/Command format version (= 1)

Byte #1:

Bits 0-1: VCL System Enable
0 = Off
1 = On
2 = On with dryer

Bit 2: 1 = Chamber A enabled, 0 = disabled

Bit 3: 1 = Chamber A dry air convey, 0 = ambient air convey

Bit 4: 1 = Chamber B enabled, 0 = disabled

Bit 5: 1 = Chamber B dry air convey, 0 = ambient air convey

Bit 6: 1 = spare

Bit 7: 1 = spare

Bytes #2-3: Chamber A load time, in seconds (2-byte unsigned integer)

Bytes #4-5: Chamber A dump time, in seconds (2-byte unsigned integer)

Bytes #6-7: Chamber B load time, in seconds (2-byte unsigned integer)

Bytes #8-9: Chamber B dump time, in seconds (2-byte unsigned integer)